METRIC

DOD-C-24529(SH)
25 March 1976
SUPERSEDING
(See 6.6)

MILITARY SPECIFICATION

CHARGERS, BATTERY, LIFT TRUCK AND

PALLET TRANSPORTER BATTERY SERVICE (METRIC)

This specification is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers semiconductor (silicon) battery chargers for charging lead-acid storage batteries covered in W-B-133, type I, and used for propulsion and operation of industrial type lift trucks and pallet transporters.

1.2 Classification. Battery chargers shall be of the following types, as specified (see 6.1.1):

Type I - For charging 6-cell (12 volts nominal), 600 to 900 ampere-hour (Ah) batteries.
Type II - For charging 12-cell (24 volts nominal), 600 to 900 Ah batteries.
Type III - For charging 18-cell (36 volts nominal), 600 to 900 Ah batteries.
Type IV - For charging 12-cell or 18-cell (24 or 36 volts nominal), 600 to 900 Ah batteries.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-P-35 - Passivation Treatments for Corrosion-Resisting Steel.
QQ-P-416 - Plating, Cadmium (Electrodeposited).
QQ-Z-325 - Zinc Coating, Electrodeposited, Requirements for.
TT-P-664 - Primer Coating, Synthetic, Rust-Inhibiting, Lacquer-Resisting.

MILITARY

MIL-C-25 - Capacitor, Fixed, Paper-Dielectric, Direct-Current (Hermetically Sealed in Metallic Cases), General Specification for.
MIL-C-62 - Capacitor, Fixed, Electrolytic (DC, Aluminum, Dry Electrolyte, Polarized), General Specification for.
MIL-S-901 - Shock Tests, H.I. (High Impact), Shipboard Machinery, Equipment and Systems, Requirements for.
MIL-C-915 - Cable and Cord, Electrical, For Shipboard Use, General Specification for.
MIL-C-915/6 - Cable, Electrical, 600 Volts, Types SHOF, DHOF, THOF, and PHOP.
MIL-C-915/9 - Cable, Electrical, 600 Volts, Type JAS.
MIL-E-917 - Electric Power Equipment, Basic Requirements for (Naval Shipboard Use).
MIL-D-1000 - Drawings, Engineering and Associated Lists.
MIL-I-1361 - Instrument, Auxiliaries, Electrical Measuring: Shunts, Resistors, and Transformers.
MIL-R-2033 - Relays for Naval Shipboard Electrical Service.
MIL-E-2036 - Enclosures for Electric and Electronic Equipment, Naval Shipboard.
MIL-C-2212 - Controllers, Electric Motor, AC or DC, and Associated Switching Devices, Naval Shipboard.

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MIL-L-3661 - Lampholders, Indicating Lights, Indicator-Light Housings, and Indicator-Light Lenses, General Specification for.
MIL-C-3965 - Capacitor, Fixed, Electrolytic, (Nonsolid Electrolyte), Tantalum, General Specification for.
MIL-M-10304 - Meters, Electrical Indicating, Panel Type, Ruggedized, General Specification for.
MIL-P-15024 - Plates, Tags and Bands For Identification of Equipment.
MIL-P-15024/5 - Plates, Identification.
MIL-M-15071 - Manuals, Technical: Equipments and Systems Content Requirements for.
MIL-P-15137 - Provisioning Technical Documentation for Repair Parts For Electrical and Mechanical Equipment (Naval Shipboard Use).
MIL-S-15291 - Switches, Rotary, Snap Action.
MIL-P-15328 - Primer (Wash), Pretreatment, Blue (Formula No. 117-B For Metals).
MIL-O-16485 - Ohmmeters, Insulation-Resistance-Indicating, Portable.
MIL-E-17555 - Electronic and Electrical Equipment, Accessories, and Repair Parts; Packaging and Packing of.
MIL-R-19523 - Relays, Control, Naval Shipboard.
MIL-C-19978 - Capacitors, Fixed, Plastic (or Paper-Plastic) Dielectric (Hermetically Sealed in Metal, Ceramic, or Glass Cases), Established and Non-Established Reliability, General Specification for.
MIL-F-21346 - Fuseholders, Block, Plug and Shroud Type and Associated Fuse Clips, General Specification for.
MIL-M-38510 - Microcircuits, General Specification for.
MIL-C-39003 - Capacitors, Fixed, Electrolytic (Solid Electrolyte), Tantalum, Established Reliability, General Specification for.
MIL-T-55164 - Terminal Boards, Molded, Barrier, Screw and Stud Types, and Associated Accessories, General Specification for.

STANDARDS

MILITARY

MIL-STD-108 - Definitions of and Basic Requirements for Enclosures for Electric and Electronic Equipment.
MIL-STD-167-1 - Mechanical Vibrations of Shipboard Equipment (Type I - Environmental and Type II Internally Excited).
MIL-STD-461 - Electromagnetic Interference Characteristics Requirements for Equipment.
MIL-STD-470 - Maintainability Program Requirements (For Systems and Equipments).
MIL-STD-471 - Maintainability Demonstration.
MIL-STD-701 - Lists of Standard Semiconductor Devices.
MIL-STD-740 - Airborne and Structureborne Noise Measurements and Acceptance Criteria of Shipboard Equipment.
MIL-STD-785 - Reliability Program for Systems and Equipment Development and Production.
MIL-STD-883 - Test Methods and Procedures for Microelectronics.
MIL-STD-1310 - Shipboard Bonding, Grounding and other Techniques for Electromagnetic Compatibility and Safety.
MIL-STD-1472 - Human Engineering Design Criteria For Military Systems, Equipment and Facilities.

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DRAWINGS

MILITARY

NAVSHIPS 9000-S6202-74228 - Fuse Holder, Double Dead Front Blown Fuse Indicator Type FHL10G, 30 Amp, 120V, 1-1/4 Inch by 1/4 Inch Fuse.

NAVSHIPS 9000-S6202-74229 - Fuse Holder, Single for 1-1/4 Inch by 1/4 Inch Fuses with Blown Fuse Indicator, Type FHL11G.

NAVSHIPS 9000-S6202-74230 - Fuse Holder, Single Dead Front, Blown Fuse Indicator Type FHL12G, 450V, 1-3/32 Inch by 1-1/2 Inch Fuses.

PUBLICATION

MILITARY

NAVSHIPS 0967-LP-312-8010 - Maintainability Design Criteria Handbook for Designers of Shipboard Electronic Equipment.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

Y32.16 - Reference Designations for Electrical and Electronic Parts and Equipments.

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A386 - Zinc-Coating (Hot-Dip) on Assembled Steel Products.

B6 - Zinc Metal (Slab Zinc).

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Qualification. The battery chargers furnished under this specification shall be products which are qualified for listing on the applicable qualified products list at the time set for opening of bids (see 4.3 and 6.3).

3.2 General requirements. The equipment shall conform to the requirements of MIL-E-917, except as specified herein. Where a requirement of this specification conflicts with a requirement of MIL-E-917, the requirement of this specification shall govern.

3.2.1 Electrical rating and performance.

3.2.1.1 A.c. input transients. When the alternating current (a.c.) input supply voltage suffers a momentary (total duration up to 2.0 seconds) rise or drop of up to 18 percent of the nominal value the charger shall:

(a) Be affected in neither calibration or adjustment, nor shall the life of any component be shortened.

(b) Suffer a change of output voltage of a value not to exceed that of the a.c. input, percentage-wise. For example, if the a.c. voltage changes by 10 percent, the direct current (d.c.) output voltage shall not change by more than 10 percent.

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3.2.1.2 Phase balance and grounding. Equipment shall operate from a 3-phase, 3-wire, ungrounded power source. Input supply current shall be balanced to the extent that the current in any wire shall not differ by more than 10 percent from the average current in the three wires. The equipment design shall not establish a ground or require grounding, either directly or by means of filter capacitors, of the power source.

3.2.1.3 D.c. output characteristics. Equipment shall operate as specified with variation of voltage and ambient temperature as specified in 3.2.1.1 and 3.2.2.2. This shall include any special output performance requirements such as voltage regulation and adjustment, current regulation and adjustment, current limiting, current overload, transient performance, and ripple voltage content. The battery charger output shall have the following characteristics and capabilities:

(a) Maximum charging time. Each battery charger shall be cabable of charging batteries with capacities between 600 and 900 Ah (6-hour discharge rate) to a complete charge in no more than 8.5 hours.
(b) Current adjustment. If adjustment of the maximum output current is required for batteries of different Ah capacities, such adjustment shall be made by means of a selector switch. The adjustment selector switch shall be provided with not more than four setting positions. Each current setting shall be clearly marked to indicate the maximum current and the related range of battery Ah capacities.
(c) Temperature rise limitation. The charging voltage and current output shall be automatically regulated to prevent the battery temperature rise due to charging (and excluding battery temperature changes due to variations in ambient temperature) from exceeding 9⅛C.
(d) Temperature compensation. The regulated voltage and current output shall be automatically compensated for variations in the ambient temperature to maintain control of battery temperature rise. This may be accomplished by using a temperature sensing element placed at a position in the charger enclosure which is substantially unaffected by the temperature rise within the charger.
(e) Battery gassing voltage. The charging voltage shall be limited during high rate charging (when the charging current is 75 amperes or more) to a value lower than the battery gassing voltage.
(f) Full charge current limit. At or before the end of the nominal 8-hour charging period (when the battery has reached full charge), the output voltage shall be automatically regulated to such value as necessary to limit the charging current to a value in amperes not exceeding 1/60 of the Ah rating of the battery being charged and such that the charger may remain connected to the battery for at least 16 hours after full charge without resulting in any discharge, and without overcharging the battery to a level detrimental to the design life of the battery.
(g) Output cable voltage drop compensation. Automatic voltage and current control shall compensate for voltage drop in the output cable (see 3.2.6.10). Compensation may be accomplished by use of a voltage compounding control matched to the resistance of the output cable and sensitive to the output current or by use of voltage sensing leads from the charger enclosure to the charging connector.

3.2.1.3.1 Output protection. The charger circuitry shall be designed so as to not be damaged by either of the following:

(a) Disconnecting the output cable from the battery while charging is in progress.
(b) Turning on the charger input power when the output cable is not connected to a battery.

3.2.1.3.2 Remote sensing. Where remote sensing is required for battery charger operation, the current drawn by such a sensing circuit shall not exceed 0.010 ampere. (See 3.2.1.8 for required fail-safe features associated with remote sensing circuits.)

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3.2.1.3.3 Reversed battery. Battery charger and operator shall sustain no damage/injury when the charger is connected with incorrect polarity to any battery of up to 900 Ah. This protection may be provided by one or more fuses.

3.2.1.3.4 Misoperation. Battery charger shall sustain no damage and operator shall not be exposed to hazards if the charger is connected to any battery of up to 900 Ah, regardless of setting of control switches, and regardless of whether the a.c. input is energized.

3.2.1.4 Current rating. The continuous load current rating of the battery charger shall be 200 amperes furnished to a completely discharged 900 Ah battery (see 6.4.2) during the first hour of charging. This continuous rating shall be indicated on the equipment identification plate.

3.2.1.5 Duty. Equipment shall be designed to operate continuously at the specified current rating.

3.2.1.6 Type of load. The type of load shall be battery charging unless otherwise specified herein. Battery charging loads shall be considered to be lead-acid type batteries. The equipment shall provide the required output voltage (see 3.2.1.3) and current when connected to the above type of load, without hunting or oscillation of output voltage. For purposes of this specification, oscillation shall be defined as any sustained variation of output voltage due to interaction between battery charger and the battery being charged. The chargers shall also be capable of supplying 200 amperes to a purely resistive load.

3.2.1.7 Ripple voltage. The maximum allowable percent ripple voltage (see 6.4.6) in the output shall be 2 percent when connected to a battery with capacity between 600 and 900 Ah. The percent of output ripple voltage shall not be exceeded regardless of the charge condition of the battery or the mode of operation (voltage or current control) of the charger. This percent of output ripple voltage shall also not be exceeded when the charger is connected to resistive loads and delivering up to rated current (see 3.2.1.4). The meaning of percent value shall be as defined herein (see 6.4.7). The wave form of the a.c. component of the output voltage shall be such that the peak-to-peak value of ripple voltage shall not exceed three times the root mean square (rms) value when both quantities are measured in accordance with 4.6.5.4.

3.2.1.8 Circuit and component protection means. The following safety and fail-safe design features shall be provided in the battery chargers:

(a) Circuits using adjustable devices shall be arranged so that no damage to any component can be caused by excessive current or voltage due to any possible setting of the adjustable device. For example, where a series resistor is adjusted to control current through a circuit element, either sufficient rating of all circuit elements, or a fixed current-limiting resistor shall be used to prevent damage to circuit elements if the adjustable resistor is set at minimum value.
(b) Battery chargers using voltage sensing leads for remote voltage regulation shall be so circuited that voltage regulation at the enclosure terminals of the charger is automatically provided if the sensing lead circuit is broken.
(c) Battery chargers shall be provided with fuses for protection against overload and short circuit on output terminals.
(d) Battery chargers shall be provided with fuses to protect surviving semiconductor rectifiers from damage when one rectifier fails by becoming a short circuit.
(e) Filter capacitor circuits shall be fused to protect the battery chargers in the event of a short-circuited capacitor. Capacitors for filtering of power circuits shall be connected in at least two independently fused circuits.

3.2.1.9 Efficiency. Design of equipment shall be such as to give the maximum practical efficiency without deviation from other requirements herein.

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3.2.2 Environmental requirements.

3.2.2.1 Mechanical. Mechanical requirements shall be as follows:

(a) Shock. The battery charger shall be designed such that it is capable of passing the shock requirements specified in MIL-S-901 for grade A, class 1 equipment.
(b) Vibration. The battery charger shall be designed such that no damage will occur or malfunction be caused either by internally excited vibrations, or by the environmental vibrations specified in MIL-STD-167-1 for frequencies up to and including 33 hertz (Hz).
(c) Inclined operation. Equipment shall have no change in operation when inclined at any angle from zero to 45 degrees from the vertical in any direction.

3.2.2.2 Ambient temperature. The rating of equipment shall be based upon an ambient temperature range of 0°C to 50°C. The specified ambient is the maximum ambient rating and shall always include the range of ambient down to 0°C.

3.2.2.3 Relative humidity. Equipment whether or not energized, shall withstand 100 percent relative humidity for a period of 168 hours at room temperature plus or minus 10°C. During this period the value of insulation resistance between transformer and reactor windings and ground and between insulated windings of the same transformer or reactor shall be not lower than the "minimum value curve" shown on figure 1. The value of dissipation factor for the transformers and reactors shall not exceed the "maximum value curve" shown on figure 2. Following humidification there shall be no evidence of material damage or excessive corrosion of any components or hardware. The electrical performance shall not be affected by humidification.

3.2.3 Airborne and structureborne noise.

3.2.3.1 Airborne noise. Airborne noise generated by the battery chargers under any operating conditions shall not exceed the limits of grade B of MIL-STD-740.

3.2.3.2 Structureborne noise. Structureborne noise shall not exceed the limits of type 3 equipment of MIL-STD-740.

3.2.3.3 Use of resilient mounting of battery charger components. The preferred method of equipment noise reduction shall be by design and construction of circuits and components used therein. Where components are mounted in the battery charger assembly by resilient mounting means (sound isolators), exposed metal parts of such components (for example, transformer cores) shall be connected to the equipment frame by a grounding lead, to reduce potential electric shock hazard.

3.2.4 Electromagnetic interference (EMI). The EMI requirements for the battery chargers shall conform to MIL-STD-461, equipment class II A, procedure CE01.

3.2.5 Material requirements.

3.2.5.1 Insulation.

3.2.5.1.1 Insulation resistance. The insulation resistance of circuits shall be not less than 10 megohms when measured with circuits cold.

3.2.5.1.1.1 Design and construction of battery charger components shall be such that a 500 volt (high internal impedance ohmmeter) insulation resistance measurement shall not damage any components under the following conditions:

(a) Output (and remote sensing if used) terminals connected together.
(b) Ohmmeter connected between those terminals and ground.
(c) No insulation failure exists, that is, insulation resistance is above 10 megohms.
(d) Input terminals (3 for 3-phase equipment) connected together.
(e) Conditions (b) and (c).

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3.2.5.1.2 Voltage withstand. Equipment shall be so designed and constructed as to withstand for a period of 1 minute a test voltage case to ground as follows:

A.c. (rms) input or d.c. output voltage	Test voltage (rms)
60 and under	600
61 to 90	900
90 to 600	1000 plus twice rated
Over 600	2000 plus 2-1/2 times rated

3.2.5.2 Painting. Parts to be painted shall consist of those not having a corrosion-resisting treatment or not fabricated of corrosion-resisting materials of the types specified in MIL-E-917 and shall include exterior and interior surfaces of the enclosure. Painting shall be in accordance with MIL-E-917 except that only one coat of gray enamel shall be applied. Marks or scratches from handling during fabrication and testing shall be repainted by either complete repainting of equipment or by touch-up method.

3.2.5.3 Corrosion protection. Brass, copper, 300-series corrosion-resisting steel and galvanized steel are considered corrosion-resisting materials for this application. When the corrosion-resistance of the 300-series corrosion-resisting steel is degraded by fabrication processes, it shall be restored by heat treatment and appropriate passivation treatment.

3.2.5.3.1 Parts fabricated from other than corrosion-resisting materials shall be protected against corrosion after fabrication with chemical treatment, electrolytic processes, electroplating or specified paints. The following methods, while not restrictive, are considered corrosion protection methods when properly applied:

(a) Hot-dipped galvanized in accordance with ASTM A386 with the spelter conforming to grade 5 of ASTM B6.
(b) Electroplating with zinc in accordance with QQ-Z-325, type II, class 1.
(c) Electroplating with cadmium in accordance with QQ-P-416, type II, class 1.
(d) Phosphate or chromate treatment, or a coating of primer in accordance with MIL-P-15328 followed by two coats of primer conforming to TT-P-664.

3.2.5.3.2 Bolts, nuts, studs, screws and such fastenings or fittings as may be used shall be of corrosion-resisting material passivated in accordance with QQ-P-35, or of a material treated in a manner to render it adequately resistant to corrosion.

3.2.5.4 Dissimilar metals. To prevent destructive electrolysis, direct contact of electrolytically dissimilar materials shall be avoided. Where dissimilar metal contacts can not be avoided, care shall be exercised in material selection to prevent possible galvanic corrosion of the anodic material during operation or cleaning of the battery charger.

3.2.6 Components.

3.2.6.1 Semiconductor devices. Semiconductor devices shall be silicon types and shall be selected from MIL-STD-701. For applications which require semiconductor components of ratings other than those listed in MIL-STD-701, the supplier shall obtain components as required by MIL-E-917. Application of semiconductor devices shall be as specified in MIL-E-917. Transistors shall not be used as series regulating elements in output power circuits. Operational amplifiers shall be as specified in MIL-STD-883 and MIL-M-38510.

3.2.6.2 Transformers, reactors, and inductors. Transformers, reactors, and inductors shall conform to MIL-E-917. Transformers shall have electrically isolated windings (primary and secondary). Neither auto-transformers nor open-delta connected transformers shall be used.

3.2.6.3 Meters and meter accessories. Meters shall be furnished as an integral part of the equipment. Meters shall be understood to include any necessary meter accessories. Meters shall be in accordance with MIL-M-10304 (with external zero adjustor). Meter accessories (shunts, resistors, and similar items) shall be in accordance with MIL-I-1361. Meter range shall be as appropriate for the application to give approximately two-thirds full scale deflection at full load conditions. A voltmeter and ammeter shall be provided for measuring

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the total output current and output voltage. Meters shall be ruggedized (HI shockproof), 90 millimeter (mm) round panel type in accordance with MIL-M-10304. Ammeters shall operate from a 50-millivolt external shunt. The full scale range of the meters shall be as shown in table I.

Table I - Meter ranges.

Charger type	Ammeter	Voltmeter
	(Amperes)	
I	0-300	0-20
II	0-300	0-50
III	0-300	0-50
IV	0-300	0-50

3.2.6.4 Capacitors. Capacitors in accordance with MIL-C-3965 (wet tantalum) and MIL-C-39003 (solid tantalum) are preferred. Capacitors in accordance with MIL-C-19978 (plastic dielectric), MIL-C-25 (paper dielectric) or MIL-C-62 (electrolytic) are acceptable. Where the required capacitance is too large to be covered by MIL-C-62, it is permissible to use special sizes provided construction is similar to MIL-C-62.

3.2.6.5 Contactors and relays. Contactors, relays and other contact-making control devices shall be in accordance with MIL-C-2212, MIL-R-19523, or MIL-E-2033, as applicable.

3.2.6.6 Switches.

3.2.6.6.1 Power switch. For on-off operation of the charger at the a.c. input circuit, a contactor in accordance with 3.2.6.5 shall be used. The contactor shall provide automatic protection to the charger from circuit overloads and from excessive temperature rise within the enclosure (see 3.3.1.5).

3.2.6.6.2 Selector switch. A selector switch in accordance with MIL-S-15291 shall be used for voltage selection on type IV chargers.

3.2.6.7 Terminal boards. Terminal boards shall conform to MIL-T-55164 (with insulation in accordance with MIL-E-917) except where the terminal board is integral with and considered a part of another component (for example, contactors specified by MIL-C-2212).

3.2.6.8 Power on indicator light. Each battery charger shall be provided with an indicator light in accordance with MIL-L-3661 and shall have push-to-test provision. The indicator light shall be connected to indicate when the battery charger is energized. The indicator light shall be mounted on the front exterior of the battery charger enclosure.

3.2.6.9 Fuse and fuseholder. In addition to the overload protection provided at the a.c. input circuit, overload protection shall be provided in the d.c. load circuit by use of a fuse in one of the load lines. The fuse shall also serve to protect charger components in the event of reversed polarity battery connections.

3.2.6.9.1 Fuses. Fuses shall be provided for protection as follows:

(a) Failure of a power-handling semiconductor device (for example, shorted silicon rectifier). Surviving semiconductor devices shall be protected.
(b) Failure (shorting) or excess leakage of electrolytic filter capacitors of output circuit. The filter capacitors shall be connected and fused in at least two parallel circuits.
(c) Slow blow fuse shall be used except where fast-acting characteristics are required.
(d) An indicator light shall be provided for each fuse which is not usable in an indicator-type fuseholder. The indicator light shall be located inside the battery charger enclosure and shall be energized by a detector circuit when the related fuse has blown.
(e) All fuses shall be in accordance with MIL-E-917 except where special fuse characteristics are required and not covered by MIL-E-917.

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3.2.6.9.2 Fuseholders (with indicators). Fuseholders, with blown fuse indicators in accordance with Drawings 9000-S6202-74228, 9000-S6202-74229, and 9000-S6202-74230, shall be provided for fuses used for protection of surviving semiconductor devices from damage caused by failure of one or more other semiconductor devices.

3.2.6.9.3 Fuseholders (without indicators). Fuseholders, without blown fuse indicators, shall be clip types in accordance with the applicable types of MIL-F-21346 or the fuseholders required by 3.2.6.9.2 but with blanked indicator for fuses as follows:

(a) In applications where the required fuse is not usable in an indicator-type fuseholder.
(b) In filter capacitor circuits.

3.2.6.10 Charger output cabling. Each charger shall be fitted with a cable and connector assembly to connect the battery charger output terminals to the battery connector.

3.2.6.10.1 Connectors. The connector for the type I charger shall be type YC, receptacle half, such as Anderson Power Products catalog number 7205 or Eaton Corporation part number 0260252-00, or equal. The connector for the types II, III, and IV chargers shall be type EC, charging half, such as Anderson Power Products catalog number 5804B or Eaton Corporation part number 0169922-00, or equal. The connectors shall be fitted with strain relief devices to prevent the forces applied to the cable and connector during normal use from damaging the connections between the cable conductors and the connector contacts.

3.2.6.10.2 Cable. Cable shall conform to the requirements of MIL-C-915 and MIL-C-915/6 or MIL-C-915/9. Cable shall be 4.25 meters(m) plus or minus 50 mm in length and shall contain a minimum of two conductors of 1.28 cm^2 each. If voltage sensing leads from the charger are included in the design, cable in accordance with MIL-C-915/9 shall be used. The voltage sensing leads shall be connected to separate terminals in the charger and connected to the charging current conductors at the charging connector end.

3.2.6.11 Electrical connections. Wiring shall be neat and shall be tied or clamped in a manner that provides support and prevents chafing of the wire insulation due to vibration and shock. There shall be no splices in the wire and the connections shall be at the terminals of the components or terminal blocks.

3.2.6.11.1 External cable connections. Provision shall be made for the connection of the power source cable and the charger output cable to terminal boards within the charger enclosure. Terminal boards shall be accessible through the front access door opening.

3.2.6.11.2 Cable entrance. A blank, gasketed plate shall be provided in the left side panel of the enclosure to permit the drilling of a cable entrance by the installing activity. The charger output cable shall enter the enclosure through a stuffing tube located in the right side panel of the enclosure, at a position compatible with the location of the cable rack and the cable flexibility.

3.2.6.12 Miscellaneous component parts. To best meet equipment performance requirements, it may occasionally be desirable to make use of component parts not covered herein by referenced specifications. These parts shall be considered miscellaneous parts and may include, but may not be limited to special temperature controls or compensating devices and special switches. Miscellaneous component parts shall conform to MIL-E-917 insofar as practicable. To determine the need for and suitability of miscellaneous component parts, the manufacturer shall furnish technical data justifying its use and suitability and such other related data as may be requested by the command or agency concerned. Supplemental tests may be required on miscellaneous component parts to confirm suitability. These tests shall be conducted by the manufacturer as requested by the command or agency concerned.

3.2.6.13 Prohibited components. The following parts shall not be used:

(a) Motors.
(b) Variable transformers (typically "Variacs").
(c) Batteries.
(d) Meter relays.
(e) Electron tubes or other thermionic devices.

3.2.6.14 Prohibited materials. Materials indicated as prohibited by MIL-E-917 shall not be used.

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3.3 Detail requirements.

3.3.1 Design and construction.

3.3.1.1 Enclosure. Battery chargers shall be furnished with enclosures in accordance with MIL-E-2036. The charger enclosure shall be dripproof protected. For purposes of this specification, a "dripproof protected enclosure" shall be defined as one which excludes drops of liquid or solid particles striking the enclosure at an angle of 45 degrees from the downward verticle, and also excludes entrance of a 13 mm diameter rod through louvers or ventilation grills. The "degree of exclusion of drops of liquid or solid particles" and the "definition of failure to meet these requirements" as specified by MIL-STD-108 shall apply. Enclosures and framework shall be constructed of steel, aluminum, or both, and shall be of the dead-front type. The enclosure shall be provided with a hinged front access door. A latch shall be provided to hold the door open despite motion of the ship in which the battery charger is installed. A cable rack shall be provided on the side of the charger enclosure for storage of the charger output cable. The manual "on-off" power switch, voltage selector switch, and the meters shall be mounted on the door. Adjustments, fuses, and test points shall be readily accessible inside the front access door.

3.3.1.2 Mounting. Battery chargers shall be designed for deck mounting. Mounting provisions shall be arranged for easy access and adjustment of mounting screws or bolts without use of special tools and without disassembly of equipment. Special tools are defined as those tools not listed in the Federal Supply Catalog (copies of this catalog may be consulted in the office of the Defense Contract Administration Service (DCAS)). Battery chargers shall have mounting feet or runners of sufficient elevation to provide easy installation and to prevent blockage of any bottom ventilation openings due to accumulation of dust and debris. Provision for bonding strap in accordance with MIL-STD-1310 shall be provided.

3.3.1.3 Lifting attachments. Battery chargers shall be provided with eye-bolts or equivalent means for lifting and handling. The lifting attachments shall be so located that when lifting vertically the battery charger does not tilt more than 5 degrees from the vertical.

3.3.1.4 Size and mass. Battery chargers shall not exceed the following size and mass limits:

Width - 675 mm
Height - 965 mm
Depth - 660 mm
Mass - 306 kilograms (kg)

Battery chargers shall be designed for the minimum values consistent with operation and service required. Dimensions shall exclude extensions due to lifting eye-bolts but shall include extensions due to mounting supports, dripshields, ventilation louvers, and protruding components such as meters, indicator lights, switch handles, and similar items. Measurements of weights shall include all material (assemblies and subassemblies) of the battery chargers.

3.3.1.5 Ventilation. Ventilation shall be natural draft. The dimensions of the equipment shall include any necessary allowance for air intake and discharge. Natural draft cooling shall be designed to give maximum cooling of components with minimum size and mass. This shall be accomplished by construction of enclosure and arrangement of components to give a chimney effect natural draft with height of equipment the maximum dimension.

3.3.1.5.1 Ventilation failure protection. Protection shall be provided to avoid equipment damage from accumulation of heat upon failure of natural draft ventilation. Such protection shall operate to automatically de-energize the equipment at the a.c. input and the d.c. output to prevent reverse current flow from discharging the battery when the a.c. input is off. Upon failure of the ventilating system, the automatic protection devices shall de-energize the equipment before any semiconductor devices exceed their maximum allowable operating temperature. The following devices, listed in order of preference, may be used to accomplish the above protection function:

(a) Temperature-sensitive switch, located adjacent to a heat source, the heat from which is normally removed by the properly operating ventilating system. Upon failure of natural draft ventilation, the heat accumulates and actuates the switch.
(b) Temperature-sensitive switch, mounted in intimate contact with a heat sink of a power-handling semiconductor device.

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3.3.1.6 Arrangement of components. Components ordinarily used in the operation of the equipment shall be operable without opening any access doors. These components shall typically include on-off contactors, meters, indicator lights, selector switches, and voltage adjusting devices. The following components shall be accessible for adjustment, replacement, and test by opening a front hinged access door. Access to these components shall not require removal of any other components.

(a) Test points or terminals (see 3.3.1.6.1).
(b) Adjustable calibration devices.
(c) Switch, relay, and contactor contacts.
(d) Encapsulated components.
(e) Semiconductor components.
(f) Fuses and manual-reset overload protection devices.
(g) Terminal blocks used for connection of external wiring.

3.3.1.6.1 Test points. Test points shall be provided for control circuits for making measurements of voltage, current, and resistance as necessary for trouble shooting and adjusting operation of the equipment. The test points shall be located and identified to provide access for measurement, while the charger is operating, without hazard to personnel.

3.3.1.6.2 Adjustable calibration devices. The design of the battery chargers shall minimize the need for adjustable calibration devices. The required adjustable devices shall be located and identified to provide access for adjustment, while the charger is operating, without hazard to personnel.

3.3.1.6.3 Circuit boards. Printed circuit boards of the plug-in type shall be provided for ease of testing, removal, and replacement of components. Access to and identification of these circuit boards shall be provided to simplify the detection, removal, and replacement of defective components without hazard to personnel. Printed circuit boards shall conform to the requirements of MIL-E-917.

3.3.1.7 Reliability and maintainability. The principle of maximum reliability is paramount and no compromise of this principle shall be made with any other basic requirement of design. It is the intention of this specification to obtain equipment of such design that it will have an operating life of at least 175,200 hours (20 years). The basis for design of replaceable parts shall be an equivalent of 3 years of ship operation (approximately 21,000 hours) before replacement is necessary.

3.3.1.7.1 The designer shall take cognizance of the conditions under which the equipment will be maintained and repaired on shipboard, and of the fact that the personnel responsible for maintenance and repair may not be seasoned mechanics. Human engineering design criteria and principles shall be applied in the design of the battery charger so as to achieve safe, reliable, and effective performance by the operator and maintenance personnel, and to optimize personnel skill requirements. MIL-STD-1472 shall be utilized as guidelines in applying human engineering design criteria.

3.3.1.7.2 Reliability. The supplier shall establish and maintain an effective reliability program in accordance with MIL-STD-785 (see 6.2) and shall include the following:

(a) Design reviews. The reliability assurance program shall include provisions for the reliability review and evaluation of design as an integral part of the supplier's engineering design procedures. Design or engineering change occurring during development or production shall be subjected to comparable review procedures.
(b) Production control and monitoring. The reliability assurance program shall provide an economical and effective system of production control and monitoring to assure that reliability achieved in design is maintained during production (see 4.1.1).
(c) Subcontractor and vendor reliability. The reliability assurance program shall include provisions to assure subcontractor and vendor selection and performance consistent with the reliability requirements of the contract and applicable portions of this specification.
(d) Reliability analysis. The supplier shall analyze those factors affecting reliability. The reliability analysis shall include, but shall not be limited to the following:
    (1) List of those parts which experience and judgment show are subject to wear, material deterioration, and service failures.
    (2) Specific design features employed to attain the required

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service life of the parts with due consideration of shipboard environment and resultant conditions. Some suggested design features are choice of materials, hardness surface finishes, fits, clearances, fastenings, equipment protection fail-safe features, reparability and accessibility.

(3) Show by calculation or other means that the design does in fact fulfill the design requirements with the criteria chosen. Perform a reliability prediction (see 4.7.1).

(4) Preventive maintenance and servicing requirements necessary to the achievement of reliable equipment. Any unusual steps or precautions necessary in carrying out maintenance and servicing requirements shall be pointed out.

(e) Failure reporting, analysis, and feedback. The reliability assurance program shall incorporate a formalized system for recording, collecting, and analyzing all failures that occur during testing, installation, and operation through the tenure of the contract. Analysis shall be fed back to the supplier's engineering management and production activities on a timely basis. Failure reports received from using activity shall be integrated into this program for trouble analysis and for experience consideration for future design review.

3.3.1.7.3 Maintainability. The manufacturer shall establish and conduct a maintainability program in accordance with MIL-STD-470 (see 6.2). The battery charger shall be designed so that it can be maintained by easy replacement of removable parts or modules. The design shall conform to the design-for-maintainability requirements of MIL-STD-470 (see 6.2) and NAVSHIPS 0967-LP-312-8010. The equipment repair time (ERT) shall be 1 hour and shall be demonstrated in accordance with the test method of MIL-STD-471 for test on the median ERT (see 6.2). The construction of the battery charger (see 3.2) shall be such that wiring, terminals, and electrical connections shall be accessible for servicing and for test purposes without requiring the removal of a part or an assembly from the unit enclosure.

3.3.1.7.3.1 Test equipment. The battery charger shall be capable of being maintained, alined, calibrated, and repaired with the aid of the following test equipment:

(a) Clamp-on volt ammeter, 0 to 600 amperes and 0 to 600 volts, a.c., Weston model 633, or equal.
(b) Multimeter, 0 to 1000 volts, d.c. at 20,000 ohm/volt, 0 to 1000 volt a.c. at 5,000 ohm/volt, 0 to 10 amperes d.c. and 0 to 20 megohm, Simpson model 260, or equal.
(c) Oscilloscope, 500 kilohertz (kHz), single trace, 0.05 to 20 volt/div, 1 megohm input impedance, Tektronix model 212, or equal.
(d) Milliammeter, 0 to 150 ma. d.c., Weston model 931-04904003, or equal.
(e) Voltmeter, 0 to 600 volts a.c. and d.c., Weston model 341-1909003, or equal.
(f) Ammeter, 0 to 300 amperes d.c., Weston model 901-2904001 with shunt, Weston model 41227, or equal.

3.3.1.7.3.1.1 A supplier desiring to utilize test equipment other than as specified herein shall submit a list of test equipment to the command or agency concerned for review.

3.4 Identification, information, and label plates. Identification and label plates shall be style II and shall conform to types A, B, D, F, or H of MIL-P-15024 and MIL-P-15024/5. Information plates shall be style VI and shall conform to types F or H of MIL-P-15024. The physical dimensions of the identification and the information plates shall be equal to, or less than, the dimensions of a size number ten plate of MIL-P-15024.

3.4.1 Identification plates. Each battery charger shall be provided with an identification plate, and the plate shall contain the following information:

(a) Nomenclature.
(b) Type (I, II, III, or IV).
(c) National stock number.
(d) Component identification number (CID).
(e) Contract or order number.
(f) Electrical characteristics (12, 24, 36, or 24/36 volts).
(g) Manufacturer's serial number.
(h) Manufacturer's name and address.

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3.4.2 Information plates. Each battery charger shall be provided with an information plate which warns personnel of any potential danger of the battery charger and specifies the safety precautions that shall be observed while operating it. The proposed information plate design shall be submitted to the procuring activity for review.

3.4.3 Label plates. Each battery charger shall be provided with a label plate which shall indicate the following:

(a) On-off switch/circuit breaker.
(b) Power on indicator.
(c) Output voltage meter.
(d) Output current meter.
(e) Voltage selector switch.
(f) Current selector switch (if required).

3.4.3.1 The size of each label plate shall be equal to or less than 75 mm by 50 mm.

3.4.4 Marking of terminal boards. Terminal boards shall be marked in accordance with MIL-T-55164.

3.4.5 Identification of equipment division by reference designation. To provide identification that will aid documentation and to facilitate maintenance, each part, subassembly or assembly of the current measuring system shall be identified by reference designations in accordance with ANSI Y32.16.

3.5 Repair parts. Unless otherwise specified (see 6.1.1), repair parts shall be furnished in accordance with MIL-P-15137 (see 6.2).

3.6 Technical data. The supplier shall prepare technical data in accordance with the data ordering documents included in the contract or order (see 6.1.2) and as specified in 3.6.1 through 3.6.4.

3.6.1 Engineering drawings. In addition to the drawing content required by the data ordering document (see 6.1.2) the unique features specified in 3.6.1.1 through 3.6.1.4 shall be included.

3.6.1.1 Drawings and certification data. Drawings shall be on sheets of the same size and shall be not smaller than 430 mm by 560 mm nor larger than 710 mm by 1020 mm. The size drawing scale shall be such that the number of sheets required shall be the minimum consistent with contents and clarity and shall not exceed three without review by the command or agency concerned. Drawings shall contain the following minimum data:

(a) List of descriptive data of the equipment including:
  (1) Complete rating showing (a) input voltage, frequency, and number of phases (b) input current, and power factor at full load (c) output voltage and current at full load and (d) efficiency at full load.
  (2) Duty.
  (3) Shock classification.
  (4) Enclosure classification.
  (5) Ventilation.
  (6) Type of load (indicate resistive load ratings as well as types and ratings of batteries).
  (7) Ambient temperature range.
  (8) Output ripple content (resistive load).
  (9) Mass of equipment (approximate).
  (10) Notation of corrosion-resistant treatment of hardware.
  (11) Special features.
(b) Finish, including method of treatment of enclosure for paintings, color and applicable specification of paint.
(c) List of material excluding items of hardware (identify repair parts).
(d) Mounting clearances for access and ventilation.
(e) Outline of equipment showing overall and principal dimensions of front view, side view, top view, and sectional views, as necessary, to show approximate mounting arrangement and location of components of the list of material. Components shall be flagged by item or piece number. Views shall show identification and information plates with data to appear thereon. Locate center of gravity.
(f) A table of insulation indicating the location, class of insulation,

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insulation material and applicable specifications and remarks.

(g) A table or list of component units such as transformers, relays, contactors, switches, fuses, semiconductors, resistors, capacitors, and similar items giving piece number, type, NSN, and rating of each. This data may be incorporated in the list of material. Ratings shall include both manufacturer's rating and the specific application rating.

(h) Sufficient detail of component locations and mounting means to aid in any disassembly likely to be needed for maintenance.

(i) Transformer and reactor data:
  (1) Core material and core or lamination form and size (including stack height). If core boxes are used, the material and size of the box shall be given.
  (2) Winding data including number of turns, taps, wire size and specification type designation, insulation, method of impregnation and treatment, and d.c. resistance at a specified temperature.
  (3) If potted, the method of potting and potting compound shall be identified.
  (4) Identification as to where used (circuit designation, for example: transformer T1).

(j) Semiconductor data:
  (1) General information including the name of manufacturer, manufacturer's identification number, circuit connection, number of devices per equipment and identifications as to where used.
  (2) Manufacturer's rating at a specified ambient temperature with a resistve load including maximum voltage, current, nominal reverse current (if applicable), duty if other than continuous, cooling if other than convection cooled and maximum operating temperature.
  (3) Supplier's application data including voltage, current, reverse current requirements (if any), duty, type of load, and ambient temperature of device with equipment at full load at rated ambient.

(k) Curves on operating characteristics of magnetic amplifiers where applicable, including the following:
  (1) Plot of load current as a function of control ampere turns (per core) for 90, 100, and 110 percent rated voltage and at a rated load and frequency.
  (2) Plot of load current as a function of control ampere turns (per core) for 95 and 105 percent rated frequency and at rated load and voltage.
  (3) Plot of the 63 percent response time as a function of circuit turns squared per ohm at rated supply voltage and frequency and rated resistance load.

Note: The control circuit turns squared per ohm at which the curves of items (k)(1) and (k)(2) are taken shall be shown and shall be the same for each condition. The same control windings shall be used for taking the data for all of the curves and shall be identified. In determining the value of turns, squared per ohm, the turns per core and the d.c. resistance of the complete control circuit including the signal source shall be used.

(l) Detailed dimensioned sketch, with terminal markings, of terminal boards for input and output connections and test points.

(m) Schematic electrical diagram arranged in simplified form. Components and windings shall be identified by appropriate symbols corresponding to their identification in the description of operation.

(n) Wiring diagram indicating markings of terminals, leads, and component parts. Each lead shall be shown by number designation. The wiring diagram shall show components and wiring in the approximate same physical location with respect to one another as mounted in the equipment.

(o) Description of operation covering manual procedures required of an operator, adjustments, and a general theory of operation.

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3.6.1.2 Arrangement of data on drawings shall be such that:

(a) When only one sheet is used, the data of items (a) through (o) of 3.6.1.1 shall be on the one sheet.
(b) When two sheets are used, the data of items (a) through (l) of 3.6.1.1 shall be contained on sheet number 1 and the remaining data items (m) through (o) of 3.6.1.1 shall be on sheet number 2.
(c) When the maximum three sheets are used, the data of items (a) through (f) of 3.6.1.1 shall be contained on sheet number 1, the data of items (g) through (l) of 3.6.1.1 shall be contained on sheet number 2 and the remaining data of items (m) through (o) of 3.6.1.1 shall be on sheet number 3.

3.6.1.3 Data, items (h) through (k) of 3.6.1.1 are not required on drawings.

3.6.1.4 Preliminary drawings. Preliminary drawings, shall contain design data, items (a) through (g) (i), (j), (l), (m), and (o) of 3.6.1.1.

3.6.2 Technical manual. The supplier shall prepare a technical manual in accordance with the data ordering document included in the contract or order (see 6.1.2).

3.6.3 Safety.

3.6.3.1 Operating hazard analyses. Analyses shall be performed to determined safety requirements for personnel, procedures, and equipment used in installation, maintenance, support, testing, transportation, storage, operations, and training during all phases of intended use as specified in the system requirements. Engineering data, procedures, and instructions developed from the engineering design and initial test programs shall be used in support of this effort. Results of these analyses shall provide the basis for:

(a) Design changes where feasible to eliminate hazards or provide safety devices, and safeguards.
(b) The warning, caution, special inspections and emergency procedures for operating and maintenance instructions including emergency action to minimize personnel injury.
(c) Identification of a hazardous period time span and actions required to preclude such hazards from occurring.
(d) Special procedures for servicing, handling, storage, and transportation.
(e) Inputs to section 8 of the service manual.

3.6.3.2 Operating hazard analysis report. The supplier shall provide an operating hazard analysis report in accordance with the data ordering document included in the contract or order (see 6.1.2).

3.6.4 Support equipment list. The supplier shall provide a support equipment list (see 3.3.1.7.3.1.1) in accordance with the data ordering document included in the contract or order (see 6.1.2).

3.7 Workmanship. Sharp edges, burrs, and other imperfections shall be removed from parts subject to contact with personnel to prevent cuts during repair and maintenance. Face of meters shall not be scratched or broken. Fasteners shall be in place and tight.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection system program plan. The supplier shall provide and maintain an inspection system program plan in accordance with the data ordering document included in the contract or order (see 6.1.2).

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4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Qualification inspection (see 4.3).
(b) Quality conformance inspection:
    (1) Routine inspection (see 4.4.1).
    (2) Selective inspection (see 4.4.2).
    (3) Periodic inspection (see 4.4.3).

4.3 Qualification tests. Qualification tests shall be conducted at a laboratory satisfactory to the Naval Ship Engineering Center (NAVSEC). Qualification tests shall consist of the examination and tests specified in table II and shall be conducted in the order listed.

4.3.1 Samples for qualification tests. One battery charger of each type for which qualification is desired shall be subjected to the qualification tests specified in table II. Qualification approval will be granted only for the battery charger which has been submitted and which has successfully passed the qualification tests of table II.

4.3.2 Prior to authorization of qualification tests, the supplier shall submit detailed drawings (see 3.6.1) and a test plan to NAVSEC. Manufacture of equipment for tests shall not begin until drawings and test plan are approved by NAVSEC (except for prototype breadboard samples which may be built any time by the supplier to prove certain operating features and circuitry). The test plan shall detail the following:

(a) Method of performing each test.
(b) Test instrumentation to be used.
(c) The specifications of the test equipment (accuracy, range, and frequency response, if applicable).
(d) Diagrams showing the connections to be made to the test instruments.

Table II - Qualification, routine, selective, and periodic inspection.

Description	Qualification	Routine	Selective	Periodic	Reference paragraph	
					Requirement	Inspection
General examination	X	X	---	X	---	4.5
Mass	X	---	X	X	3.3.1.4	4.6.1
Creepage and clearance distance	X	---	---	X	MIL-E-917	4.6.2
Insulation resistance	X	X	---	X	3.2.5.1.1	4.6.3
Effectiveness of enclosure	X	---	---	X	3.3.1.1	4.6.4
D.c. output and special performance requirements	X	---	X	X	3.2.1.3	4.6.5
Heating	X	---	---	X	3.3.1.5	4.6.6
Phase balance and grounding	X	X	---	X	3.2.1.2	4.6.7
Efficiency	X	---	X	X	3.2.1.9	4.6.8
Inclined operation	X	---	X	---	3.2.2.1	4.6.9
Environmental/humidity	X	---	---	---	3.2.2.3	4.6.10
Shock	X	---	---	---	3.2.2.1	4.6.11
Vibration	X	---	---	---	3.2.2.1	4.6.12

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Table II - Qualification, routine, selective, and periodic inspection (con.).

Description	Qualification	Routine	Selective	Periodic	Reference paragraph	
					Requirement	Inspection
Voltage withstand	X	X	---	X	3.2.5.1.2	4.6.13
Airborne noise	X	---	---	---	3.2.3.1	4.6.14
Structureborne noise	X	---	---	---	3.2.3.2	4.6.14
Electromagnetic interference	X	---	---	X	3.2.4	4.6.15

4.4 Quality conformance inspection.

4.4.1 Routine inspection. Each battery charger offered for delivery shall be subjected to the routine inspection specified in table II. If any battery charger fails in any routine examination or test (see 4.7), the manufacturer shall take corrective action to enable the battery charger to conform to the requirements of this specification.

4.4.2 Selective inspection.

4.4.2.1 Definition of a lot. All battery chargers of the same type offered for delivery at one time shall be considered a lot. The lot may include the entire quantity required by one contract or it may be the production of any convenient time period.

4.4.2.2 Sampling for selective inspection. A random sample of battery chargers shall be selected from each lot in accordance with table III, and shall be subjected to the selective inspection of table II. Failure of any battery charger in the selected sample to pass any of the selective inspection shall be cause for rejection of that lot (see 4.7). If a lot is rejected due to failing the selective inspection no further lots will be accepted for inspection until the manufacturer has taken corrective measures and has demonstrated, to the satisfaction of the Government representative that these measures will enable the battery chargers to conform to the requirements of this specification.

Table III - Sampling for selective inspection.

Lot size	Sample size
1 to 5	1
6 to 8	2
9 to 10	3
11 to 15	4
16 to 40	5
41 to 65	7
66 to 110	10
111 to 180	12
181 to 300	15
301 or more	Not less than 5 percent

4.4.3 Periodic inspection. The first battery charger of each type and design offered for delivery shall be subjected to the periodic inspection of table II. A periodic inspection shall be required after any change in design affecting performance characteristics. Only one periodic inspection of a specific type and design of battery charger will be required on a contract or order. In case of subsequent contracts or orders for identical battery chargers, the periodic inspection shall be repeated if 18 months or more have lapsed between the date of the last previous periodic inspection and the date of the subsequent contract or order. However, shock and vibration tests are required only once on any specific

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type and design of battery charger. If the battery charger subjected to the periodic examination and tests fails in any examination or in any test (see 4.7), no battery charger shall be accepted for routine inspection until the manufacturer has taken corrective action on that battery charger and on every other battery charger to be offered for delivery to assure that the battery charger complies with the requirements of this specification. The examination or test failed, and any other periodic examination and test required by the Government shall be repeated to demonstrate that the corrective action will enable the battery charger to conform to the requirements of this specification. Only one re-examination and retest will be permitted. The re-examination and retest shall be made within such time after the first examination and test as stipulated by the Government. Failure to correct defects and successfully pass the re-examination and retests shall be cause for final rejection of the battery charger.

4.5 General examination. Items on major parts and complete battery chargers shall be examined as specified in 4.5.1 and 4.5.2 to determine conformance with the requirements of this specification.

4.5.1 Major parts. Examination shall be as follows:

(a) To determine conformance with the applicable specification and that the size, type, and rating are as shown on the drawing for the following items:
  (1) Resistors and rheostats.
  (2) Instruments, instrument transformers, shunts, and other associated accessories.
  (3) Laminated materials, such as glass mica glass (GMG), glass silicon glass (GSG) used for terminal boards, sub-panels, etc.
  (4) Circuit breakers and fuses.
  (5) Capacitors.
  (6) Switches.
  (7) Semiconductor devices.
  (8) Terminal boards.

(b) Materials used in the construction of reactors, magnetic amplifiers, and transformers are as shown on the drawing and comply with the material requirements as shown on the drawing.
(c) Windings of reactors, magnetic amplifiers, and transformers were given varnish impregnation in accordance with the procedures shown on the drawing.
(d) No black varnish, glyptol, paint, or other treatment was given to the insulation of reactors, magnetic amplifiers, and transformers.
(e) The resistance of windings of reactors, magnetic amplifiers, and transformers is as shown on the drawing. Ratio of turns is as specified on the drawing.
(f) The equipment enclosure is as shown on the drawing.

4.5.2 Completed battery charger. Examination shall be conducted to determine the following:

(a) Components are mounted as shown on the drawing.
(b) Parts which require servicing, repair, replacement, or periodic adjustment during the life of the equipment are readily accessible.
(c) Components are marked with identifying symbols and these symbols agree with those used on the drawing.
(d) Terminals of components and terminal boards are marked with identifying letters and numbers, or both, and this identification agrees with those used on the drawing.
(e) Wire used for interconnecting components is of the type shown on the drawing.
(f) Wiring is neatly formed into groups and laced with nonflammable glass cord and supported or clamped in a manner which will prevent chafing of the insulation due to vibration and shock.
(g) Cable clamps used are of nonflammable material.
(h) There are no splices in any of the wires.
(i) Wire groups from hinged doors and panels are formed and clamped in a manner that sharp bends do not occur with the panel or door in

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either the open or closed position.
(j) Sufficient slack in wiring is allowed so that the mass of the harness is not supported by the terminal connections and so that at least two replacements can be made if lugs are clipped off at the wire end.
(k) Cable entrance provisions are as shown on the drawing.
(l) Sufficient space is provided to bring external cables through these cable entrances, support them within the enclosure, and make connections at the appropriate terminals.
(m) Wherever wires run through holes in metal partitions or chassis, grommets are provided for mechanical protection.
(n) Wires are not bent around sharp corners which may injure the insulation.
(o) Wires are connected by either bolted or soldered connections.
(p) Bolted connections are provided with locking devices.
(r) Both ends of wires are marked with designations as shown on the drawing.
(s) Hinged doors and panels do not bind when opening.
(t) A door stop and positioning device is furnished and functions satisfactorily.
(u) Identification, information, and label plates are furnished as shown on the drawing.

4.6 Tests. At least 3 charges of a fully discharged 900 Ah battery (see 3.2.1.4) shall be conducted to determine if the performance requirements are satisfactory for qualification tests. For the routine, selective, and periodic tests, resistive loads shall be used instead of a battery. Three different resistance values shall be selected to simulate charger operation at (a) the maximum charging current, (b) the gassing voltage charging current, and (c) the full charge current limit, respectively.

4.6.1 Mass. The mass of the battery charger and the mass of the onboard repair parts shall be taken and recorded separately. The mass of the battery charger shall not exceed the value specified in 3.3.1.4.

4.6.2 Creepage and clearance distances. Creepage and clearance distances shall be demonstrated by actual measurement to be in accordance with MIL-E-917.

4.6.3 Insulation resistance. Insulation resistance shall be as specified in 3.2.5.1.1. The measurement of insulation resistance shall be made with circuits of equal voltage above ground connected together. Circuits or groups of circuits of different voltage above ground shall be tested separately. Insulation resistance shall be measured with an insulation-resistance-indicating meter conforming to type GC of MIL-O-16485. The time of test voltage application shall be not less than 60 seconds. Measurements shall be made at any convenient ambient temperature with circuits cold. The relative humidity and ambient temperature shall be recorded. The conditions specified in 3.2.5.1.1.1 shall apply.

4.6.4 Effectiveness of enclosure. The battery charger enclosure shall be tested in accordance with MIL-E-2036 for the corresponding type of enclosure except for types dripproof and dripproof protected. Dripproof and dripproof protected enclosures shall be examined to ascertain that the enclosure effectively performs its specified function in accordance with MIL-STD-108.

4.6.5 Performance tests.

4.6.5.1 Output characteristics. Battery chargers shall be tested under all conditions of voltage, frequency, ambient temperature, and load within the specified rating range. The output characteristics (see 3.2.1.3) shall remain within the specified limits without adjustment other than the initial setting. Output characteristic tests shall be conducted under the conditions specified in table IV. The first set of readings shall be taken within 2 minutes after the battery charger is energized.

(a) The load shall be a discharged lead-acid battery (see 3.2.1.4) or resistive load as specified in 4.6. Each condition of line voltage, frequency, and ambient temperature listed in table IV shall be varied in approximately equally spaced steps throughout the 8.5-hour charge period.

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Table IV - Conditions of line voltage, frequency, and ambient temperature.

Input voltage volts	Input frequency Hz	Ambient temperature °C
Rated 440	Rated 60	50
Rated +5 percent	Rated +5 percent	50
Rated +5 percent	Rated -5 percent	50
Rated -5 percent	Rated +5 percent	50
Rated -5 percent	Rated -5 percent	50
Rated +5 percent	Rated +5 percent	Approx. 23 (room)
Rated +5 percent	Rated -5 percent	Approx. 23 (room)
Rated -5 percent	Rated +5 percent	Approx. 23 (room)
Rated -5 percent	Rated -5 percent	Approx. 23 (room)
Rated	Rated	Approx. 23 (room)

(b) Procedure of regulation runs in 4.6.5.1(a) (table IV) shall be repeated for voltage settings over the range of specified adjustment. The selected voltage settings shall be at the minimum and maximum limit of specified voltage adjustment and at any other convenient voltage setting approximately half way between these extremes.

(c) Readings shall be taken and recorded (after 1 minute and every 30 minutes) on input voltage and frequency, ambient temperature, internal (electrolyte) temperature, specific gravity, output voltage and output current for all of the conditions specified in 4.6.5.1(a) and (b). The following determinations shall be made and curves plotted:

(1) During the charging period the voltage and current output shall be automatically regulated as necessary to prevent the battery temperature rise from exceeding 9°C (excluding battery temperature changes due to variations in ambient temperature).

(2) The battery gassing voltage shall not be reached during high current charging periods (when the charging current is 75 amperes or more).

(3) Following the third 8.5-hour normal charge period (when the battery has reached full charge), leave the charger connected to the battery. The output voltage shall be automatically regulated to such value as necessary to limit the charging current to a value in amperes not exceeding 1/60 of the Ah rating of the battery. This overcharge will continue for 16 hours to insure no detriment to battery design life.

(4) Checks shall be made to see that compensation is automatically made by the charger regulation for ambient temperature fluctuations affecting charge voltage.

(5) Verification shall be made of the automatic compensation for voltage drop between charger and battery based on a cable length of 4.25 m of 2-conductor number (1.28 cm^2 each) load cable.

(6) Curves shall be plotted of all measurement results versus time.

4.6.5.2 Resolution. For battery chargers in which output voltage or current is adjusted by operation of a selector switch or by connection of transformer taps in a specific manner to yield a specific output voltage or current value, tests to determine that a specific desired value of voltage or current (within specified tolerance) can be obtained shall be run. Procedure shall consist of operating selector switch(es) or of connecting transformer leads or taps to all normal positions, recording no load and full load voltage at high, low, and rated input voltage and frequency. Data shall be compared with specified performance to determine that specified resolution is obtained.

4.6.5.3 Line voltage transient test. Tests shall be made to determine conformance with the requirement of 3.2.1.1 as follows:

(a) Test conditions. Battery charger shall operate at rated output current. Test shall be run at rated input frequency and room ambient temperature.

(b) Test features. Circuit for producing transients in a.c. voltage

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shall consist of a variable a.c. supply connected through some known impedance to the battery charger. A contactor arranged to insert or remove (short) the impedance shall be provided. In all transient tests the impedances and the contactors shall be 3-phase. Recording devices shall be as specified in 4.6.5.5.

(c) Procedure. A.c. supply shall be adjusted so that the battery charger receives rated voltage. Contactor shall be energized so that series impedance is inserted or removed, as applicable to produce voltage dip or rise. The rms values of input voltage and current and the output voltage and current shall be recorded. Tests shall be in accordance with table V.

Table V - Input transient tests.

Battery charger voltage (output)	Transient magnitudes, input voltage	
	Dip	Rise
	Percent	Percent
Maximum	10 18	10 18
Minimum	10 18	10 18

4.6.5.4 Ripple voltage. Tests specified in 4.6.5.4.1 and 4.6.5.4.2 shall be conducted to determine conformance with the requirements specified in 3.2.1.7.

4.6.5.4.1 Method of measuring ripple voltage. Method of measuring ripple voltage shall be as follows:

(a) Measure effective rms value of the ripple voltage by any one of the following procedures:
  (1) By sensitive electrodynamometer type a.c. meter or thermal type meter in series with a blocking capacitor.
  (2) By vacuum tube voltmeter, calibrated in rms values, in series with a blocking capacitor.
  (3) By oscilloscope. The peak-to-peak value of the ripple voltage shall be recorded and considered to be equal to three times the rms ripple voltage.

(b) Measure average value of the pulsating d.c. output voltage with a permanent magnet moving coil meter or rectifier type meter.

(c) Calculate percent ripple:

$$\text{Percent ripple voltage} = \frac{\text{rms of ripple voltage}}{\text{average of pulsating voltage}} \times 100$$

4.6.5.4.2 Conditions for measuring ripple voltage. Measurements specified in 4.6.5.4.1 shall be made at no-load, one-half of maximum charging current and maximum charging current, at rated input voltage and with input phase voltages unbalanced as follows:

Test condition	ØA	ØB	ØC
1	Rated	Rated	Rated -3 percent
2	Rated	Rated -3 percent	Rated -3 percent
3	Rated -3 percent	Rated -3 percent	Rated
4	Rated -3 percent	Rated	Rated
5	Rated -3 percent	Rated	Rated -3 percent

4.6.5.5 Electrical test instruments. Indicating meters used for measurement of voltages and currents (input and output) during tests shall have maximum error of indication of one-quarter percent of full scale indication. Meters used for measurement of ripple content of output voltage shall have maximum error of indication the same as above, except where vacuum tube voltmeters (VTVM) are used. Where VTVM's are used for ripple measurements, the maximum error of indication shall not exceed 5 percent. Transient performance shall be recorded with oscillographs equal or similar to Heiland "Visicorder" with galvanometers of appropriate frequency response, or with oscilloscope and camera combination Tektronix, Inc., type 536, or equal.

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4.6.5.6 Reversed battery. The battery charger shall be connected to a d.c. voltage source simulating a 900 Ah battery, such that the polarity is reversed in relation to the correct charging polarity. The simulation with all possible switch combinations shall be for both a fully charged battery and a discharged battery and shall conform to the requirements of 3.2.1.3.3.

4.6.5.7 Misoperation. The battery charger shall be connected to a battery of 900 Ah with the correct switch settings and shall be tested as follows:

(a) The charger shall be energized; 1 minute after start of charge the output cable shall be disconnected from the battery while the charging is in progress.
(b) The charger shall be energized while disconnected from the battery.
(c) The charger shall not be energized and be connected to a fully charged battery.

In (a) through (c) above, no damage shall be done to the charger in conformance with the requirements of 3.2.1.3.1 and 3.2.1.3.4.

4.6.6 Heating. Heating tests shall be made under conditions equivalent to normal operating conditions (except 4.6.1.1) at maximum rated voltage, maximum rated load, and the duty specified. The test methods to be employed and the precautions to be observed shall be as specified in 4.6.6.1 through 4.6.6.4.

4.6.6.1 Assembly and mounting of the battery charger. Heating tests shall be made only with the battery charger completely assembled and mounted in the manner for which it is designed. Barriers shall be placed adjacent to the enclosure, at the manufacturer's recommended minimum clearance for adequate ventilation, to simulate shipboard space restrictions. Heating tests shall be made on coil windings (including power transformers and reactors) and rectifier cells except for those rectifier cells of control circuits which clearly do not contribute to the overall temperature rise within the battery charger enclosure.

4.6.6.1.1 Ventilation failure protection. To check the protective devices (see 3.3.1.5.1) that insure equipment deactivation for ventilation failure, the natural draft ventilation shall be mechanically blocked to determine that deactivation occurs prior to equipment damage.

4.6.6.2 Method of temperature measurements. The method of temperature measurement shall be in accordance with MIL-E-917.

4.6.6.3 Duration of test. The heating test shall be continued a minimum of 8.5 hours or until the temperature rise of components that can be observed during test have attained a steady final value by not varying more than 3 degrees over a 3-hour period.

4.6.6.4 Method of loading. The battery charger shall be loaded with rated applied input voltage and frequency at full rated output load. The type of load shall be a discharged battery (lead-acid) 600 to 900 Ah capacity or resistive load as specified in 4.6.

4.6.7 Phase balance and grounding. The battery charger shall be tested at full load for current balance in the a.c. lines. The rms value of the a.c. line current of each phase shall be measured. Readings shall also be taken to verify no grounding of power source to battery charger (see 3.2.1.2).

4.6.8 Efficiency. The battery chargers shall be loaded with a discharged battery with a capacity ranging from 600 to 900 Ah. Meters shall be inserted in the a.c. input for reading voltage, current, and wattage and also in the d.c. output for measuring voltage and current. Readings shall be recorded after one minute and each 30 minutes thereafter during the maximum 8.5 hour charge at rated input voltage and frequency. Efficiencies shall be calculated by means of the following formula:

$$\text{Percent efficiency (overall)} = \frac{\text{D.c. volts x d.c. amperes x 100}}{\text{A.c. watts input}}$$

These efficiencies shall equal or exceed the manufacturer's rated efficiencies.

4.6.9 Inclined operation. The battery charger shall be inclined at 45 degrees (see 3.2.2.1(c)) and the effect, if any, on performance noted (see 3.2.1.3).

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4.6.10 Environmental test.

4.6.10.1 General. Environmental test shall be conducted to determine:

(a) If transformers and reactors have received proper insulation treatment and if the manufacturer's treatment process is completely effective to give the desired design life.
(b) If components and hardware have received the proper corrosion-resistant treatment and if the assembly procedures of the manufacturer in any way destroy the complete effectiveness of the treatment.
(c) If any miscellaneous weaknesses exist in construction.
(d) If performance is affected by exposure to the environmental requirements specified in 3.2.2.

4.6.10.2 Procedure. Transformer and reactor windings shall be disconnected as necessary from other components to permit accurate measurements of insulation resistance and dissipation factor of the winding and ground insulation. Initial measurements of insulation resistance and dissipation factor shall be taken as well as the temperature and relative humidity of the surrounding air. The equipment shall next be conditioned for a period to arrive at a dry condition. Conditioning shall be obtained by placing the equipment in a heat chamber or drying oven for a period of 6 hours with temperature maintained at 105°C. At the end of this period the equipment shall be removed and after cooling (within 8 hours) the insulation resistance and dissipation factor shall again be measured. The equipment shall then be immediately placed in the humidity chamber for a period of 1 week (approximately 168 hours). While under humidification, daily measurements of ambient temperature, insulation resistance and dissipation factor shall be made. At the end of the seventh day humidification period and within 5 minutes after removal from the humidity chamber the insulation resistance and dissipation factor shall be measured. Following measurements the equipment shall be carefully examined for any signs of material damage and corrosion. The extent of any observed damage or corrosion shall be noted and cause determined if feasible. After examination, any disconnections or disassembly shall be reverted to original condition and the equipment shall be operated electrically to determine if the electrical performance has in any way been affected. Also following operation, insulation resistance shall be measured and recorded between isolated circuits and between circuits and ground.

4.6.10.2.1 Recommended test apparatus. Recommended test apparatus shall be as follows:

(a) The humidity chamber may be made of steel or by placing a transparent plastic film having a low moisture permeability over a steel framework. The top of the chamber shall be slanting or peaked (or otherwise arranged) so that the excess condensate does not drip on the equipment. The humidity shall be maintained at 100 percent RH by electrically heating an open pan of water within the chamber. The amount of exposed surface of the water shall be sufficient to produce minute droplets of condensate on the equipment and insulation surfaces under test. However, the amount of condensate shall be controlled so as not to produce puddles or streams of water on the insulation surfaces. The equipment door or panels shall be opened or removed to permit free access of the humidified air to all parts and surfaces.
(b) The insulation resistance measuring equipment shall be similar and equivalent in accuracy to the General Radio Megohm Bridge, type 554-B, 500 volt d.c. with a range from 0.1 to 1,000,000 megohms.
(c) The dissipation factor measuring equipment shall be similar and equivalent in accuracy to the General Radio Capacitance Test Bridge, type 1611-A, 60 Hz with a range of dissipation factor from 0 to 50 percent.

4.6.10.2.2 Methods of measurement. Methods of measurement shall be as follows:

(a) The insulation resistance measurement shall be made using a potential of 500 volts, d.c. applied for a 1 minute period. The insulation resistance shall be measured on all transformers and reactors. Measurements shall be made and recorded between each winding and ground and between insulated windings.

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(b) The dissipation factor shall be measured with the capacitance test bridge with values read directly in percent from the bridge. Dissipation factor shall be measured on all transformers and reactors. Measurements shall be made and recorded between each winding and ground and between insulated windings.

NOTE: Due to the need for exposing copper for the connections to the windings, a leakage path is produced by humidification over the various insulation surfaces between the base copper and the other parts of the equipment that are otherwise insulated from the copper circuit. Therefore, leads should be held separated as much as possible and covered with a nonwetting grease which is wiped off with a clean dry rag upon removal from humidification. Measurements shall be made either with equipment in the humidity chamber or within 5 minutes after removal of the equipment from the humidity chamber.

4.6.11 Shock. The high impact shock tests (electrical operation and mechanical damage) shall be conducted in accordance with type A of MIL-S-901.

4.6.11.1 Test features. Test features shall be as follows:

(a) General. Type A (for completely assembled apparatus) test shall be performed under light or medium weight classification, as required. In all cases the battery charger shall be checked for d.c. output and special performance requirements (see 4.6.5) following the shock test. The battery charger shall be operating at full load during the shock test. Electrical tests as required shall be made to determine satisfactory performance following or during shock test.

(b) Definition of failure to perform principal functions:
  (1) Breakage of parts, including mounting bolts. Minor chipping of parts such as plastic knobs and cases and minor distortion of parts will be permitted where such chipping or distortion cannot in any manner impair operation of the battery charger as specified.
  (2) Appreciable distortion of parts, including enclosure and framework.
  (3) A value of insulation resistance lower than that permitted by this specification (see 3.2.5.1.1).
  (4) Low voltage withstand. After shock tests the voltage withstand test shall be conducted at a voltage equal to 65 percent of the voltage specified in 3.2.5.1.2. Failure to pass this test shall be cause for rejection.
  (5) Failure to pass visual examination. The battery charger shall be carefully examined after removing all removable panels and doors to ascertain mechanical damage. When required, partial disassembly shall be performed to aid in determining possible damage.
  (6) Failure to perform the required electrical tests, during and following shock tests. No adjustment or replacement of damaged parts shall be permitted during shock test unless permitted by NAVSEC.

(c) Mounting. The battery charger shall be mounted on the shock machine in a manner simulating shipboard installations. The specific method or methods of mounting shall be determined as required by MIL-S-901.

(d) Place at which shock test will be conducted. At a Government laboratory, at the manufacturer's plant, or other testing facility satisfactory to NAVSEC.

(e) Number of battery chargers to be shock tested. One battery charger of each type shall be shock tested during qualification or periodic tests.

(f) Disposal of shock tested battery chargers.
  (1) Battery chargers which have been subjected to HI shock test under qualification tests shall be returned to the manufacturer for his post shock tests in accordance with MIL-S-901.
  (2) Battery chargers which have been subjected to HI shock test shall be accepted as production battery chargers of the contract or order only under the following conditions:
    a. That damaged parts are replaced.
    b. That post shock electrical tests show compliance with specified performance. (The extent of electrical testing, after shock, for specified operation shall

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be determined as necessary, in each instance and shall be dependent on the nature of required performance and inspection results of shock test.)

c. That an identification plate is added identifying the battery charger as "HI shock tested".
d. That the battery charger is subjected to the same guarantee by the manufacturer as other production battery chargers.
e. That not more than one shock tested battery charger of a specific type or design per contract or order is being offered for acceptance following HI shock tests.

4.6.12 Vibration. Vibration tests shall be conducted in accordance with MIL-STD-167-1, type I.

4.6.12.1 Test fixtures. The test fixtures as specified for shock tests (see 4.6.11.1) shall apply also to vibration tests except that standard fixtures may be used throughout, and 30 degree fixtures are not applicable. The battery charger selected for the vibration test shall be the same battery charger as selected for the shock tests.

4.6.13 Voltage withstand tests.

4.6.13.1 General. Voltage withstand tests shall be performed after the examination (see 4.5) and tests specified in 4.6.1 through 4.6.12 have been completed. If the insulation resistance of the windings is known to be lower than specified, due to dirt, moisture or damage to windings this shall be remedied before the application of the test voltage.

4.6.13.2 Test voltage. The frequency of the testing voltage shall be not less than 60 Hz and shall approximate a true sine wave. The value of test voltage shall be as specified in 3.2.5.1.2 and shall be applied continuously for a period of 1 minute.

4.6.13.3 Measurement of test voltage. The measurement of the voltage used in voltage withstand tests shall be made by the voltmeter method whereby the meter derives its voltage from the high-voltage circuit either directly or by means of a voltmeter coil placed in the testing transformer, or through an auxiliary ratio transformer.

4.6.13.4 Points of application. The test voltage shall be successively applied between each electric circuit and all other electrical circuits and grounded metal parts not electrically connected to it. Semiconductor components may be shorted out or disconnected for the voltage withstand test.

4.6.14 Airborne and structureborne noise. Airborne and structureborne noise measurements shall be made and recorded in accordance with MIL-STD-740 with the battery charger operating at full load at rated input voltage and frequency.

4.6.15 Electromagnetic interference. Electromagnetic interference tests shall be conducted to determine conformance with the requirements of 3.2.4.

4.6.16 Maintainability demonstration. The battery charger of each type shall be examined after testing, and the capability to maintain, disassemble, and repair the battery charger shall be demonstrated to a Government representative. The demonstration shall be conducted utilizing the recommended tools and with other than expert mechanics. The maintainability demonstration shall be in accordance with MIL-STD-471 (see 6.2) to determine conformance with 3.3.1.7.3.

4.7 Test plan and test reports. The supplier shall prepare a test plan and test reports in accordance with the data ordering documents included in the contract or order (see 6.1.2), and as specified in 4.7.1 through 4.7.3.

4.7.1 Reliability prediction. The supplier shall perform a reliability prediction and prepare a report in accordance with the data ordering document included in the contract or order (see 6.1.2).

4.7.2 Maintainability demonstration test plan. A maintainability demonstration test plan shall be prepared by the supplier in accordance with the data ordering document included in the contract or order (see 6.1.2) and submitted to NAVSEC for review before the tests commence. The plan shall define in detail the maintainability actions to be performed during the test. The test plan shall include a list of sample tasks for NAVSEC review. The list of tasks will be used by the Government, at the time of the demonstration

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to select the 20 tasks to be demonstrated in accordance with method 3 of MIL-STD-471 (see 6.2).

4.7.3 Maintainability demonstration test report. A maintainability demonstration test report shall be prepared by the supplier in accordance with the data ordering document included in the contract or order (see 6.1.2) and submitted to NAVSEC for review after completion of the maintainability demonstration.

4.8 Inspection of preparation for delivery. The preservation-packaging, packing, and marking shall be inspected for compliance with section 5 of this document.

5. PREPARATION FOR DELIVERY

(The preparation for delivery requirements specified herein apply only for direct Government procurements. For the extent of applicability of the preparation for delivery requirements of referenced documents listed in section 2, see 6.5.)

5.1 Preservation-packaging, packing, and marking. Battery chargers shall be individually preserved-packaged levels A or C; packed levels A, B, or C and marked as specified (see 6.1.1) in accordance with MIL-E-17555.

5.2 Repair parts. Repair parts shall be preserved, packaged, packed and marked for the level of shipment as specified in the contract or order, in accordance with MIL-E-17555.

5.3 Manuals. Manuals shall be packaged, packed, and marked as specified in the contract or order, in accordance with MIL-M-15071.

5.4 Drawings. Drawings shall be packaged and packed in accordance with MIL-D-1000.

6. NOTES

6.1 Ordering data.

6.1.1 Procurement requirements. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type required (see 1.2).
(c) Repair parts required (see 3.5).
(d) Level of preservation-packaging, packing, and marking required (see 5.1).

6.1.2 Contract data requirements. When this specification is used in a procurement invoking the data requirement clause of the Armed Services Procurement Regulations (ASPR) paragraph 7-104.9(n) and which incorporates a DD Form 1423 Contract Data Requirements List (CDRL), the data requirements identified below will be developed as specified in the cited Data Item Description (DID) and delivered in accordance with such CDRL. When the ASPR provisions are not invoked, the data specified below shall be delivered in accordance with the contract requirements.

	Specification paragraph	Data requirement	Service	Applicable DID	Options
(a)	3.6.1	Drawings	SH	UDI-E-23174	Categories A, E, G, and H, form 2, types II and III, or type IV
(b)	3.6.2	Technical manual	SH	DI-M-2044	MIL-M-15071, type II
(c)	3.6.3.2	Systems hazard analyses report	SH	UDI-H-23391	---
(d)	3.6.4	Support equipment list	SH	DI-V-2074	---
(e)	4.1.1	Inspection system program plan	SH	DI-R-4803	---
(f)	4.7	Test/inspection report	SH	UDI-T-23473	---
(g)	4.7	Failure/malfunction report	SH	UDI-T-23724	---
(h)	4.7.1	Reliability prediction report	SH	DI-R-2117	---
(i)	4.7.2	Maintainability demonstration test plan	SH	UDI-R-23564	---
(j)	4.7.3	Maintainability demonstration test report	SH	UDI-R-23565	---

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(Copies of DID's required by the supplier in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.)

6.2 Management control system documents. The following management control system documents should be included on DD Form 1660:

(a) MIL-STD-785 (see 3.3.1.7.2).
(b) MIL-STD-470 (see 3.3.1.7.3).
(c) MIL-STD-471 (see 3.3.1.7.3, 4.6.16, and 4.7.2).
(d) MIL-P-15137 (see 3.5).

6.3 With respect to products requiring qualification, awards will be made only for products which are at the time set for opening of bids, qualified for inclusion in applicable Qualified Products List QPL-24529 whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is the Naval Ship Engineering Center, Prince George's Center, Center Building, Hyattsville, Maryland 20782, and information pertaining to qualification of products may be obtained from that activity. Application for Qualification tests shall be made in accordance with "Provisions Governing Qualification SD-6" (see 6.3.1).

6.3.1 Copies of "Provisions Governing Qualification SD-6" may be obtained upon application to Commanding Officer, Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120.

6.4 Definitions. The definitions specified in 6.4.1 through 6.4.7 shall apply to the various technical terms wherever such terms appear in this specification.

6.4.1 Rectifying element. A rectifying element is a circuit element which conducts current effectively in one direction only.

6.4.2 Complete discharge. Complete discharge is a battery condition in which the volts per cell cannot be maintained above 1.50 at a resistive discharge load of 200 amperes with an electrolyte temperature of 25°C.

6.4.3 Complete charge. Complete charge is the battery condition following a charging period beginning with complete discharge and during which the integrated Ah input is not less than 110 percent of the rated capacity.

6.4.4 Battery charger. A battery charger is a device utilizing an a.c. source to reactivate batteries in which reversible chemical reaction occurs.

6.4.5 Full-wave rectification. Full-wave rectification is rectification in which both halves of the a.c. cycle are transmitted as unidirectional current.

6.4.6 Ripple voltage. The ripple voltage is the alternating component of a substantially unidirectional voltage.

6.4.7 Percent ripple voltage. The percent ripple voltage is the ratio in percent of the effective (rms) value of the ripple voltage in the pulsating voltage to the average value of the pulsating voltage.

6.5 Sub-contracted material and parts. The preparation for delivery requirements of referenced documents listed in section 2 do not apply when material and parts are procured by the supplier for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.6 Supersession data. This specification replaces types I, II, and III battery chargers formerly covered in MIL-P-15736/3(SHIPS), which was canceled 15 October 1974 and adds type IV (see 1.2).

Preparing activity:
Navy - SH
Project 6130-N164)

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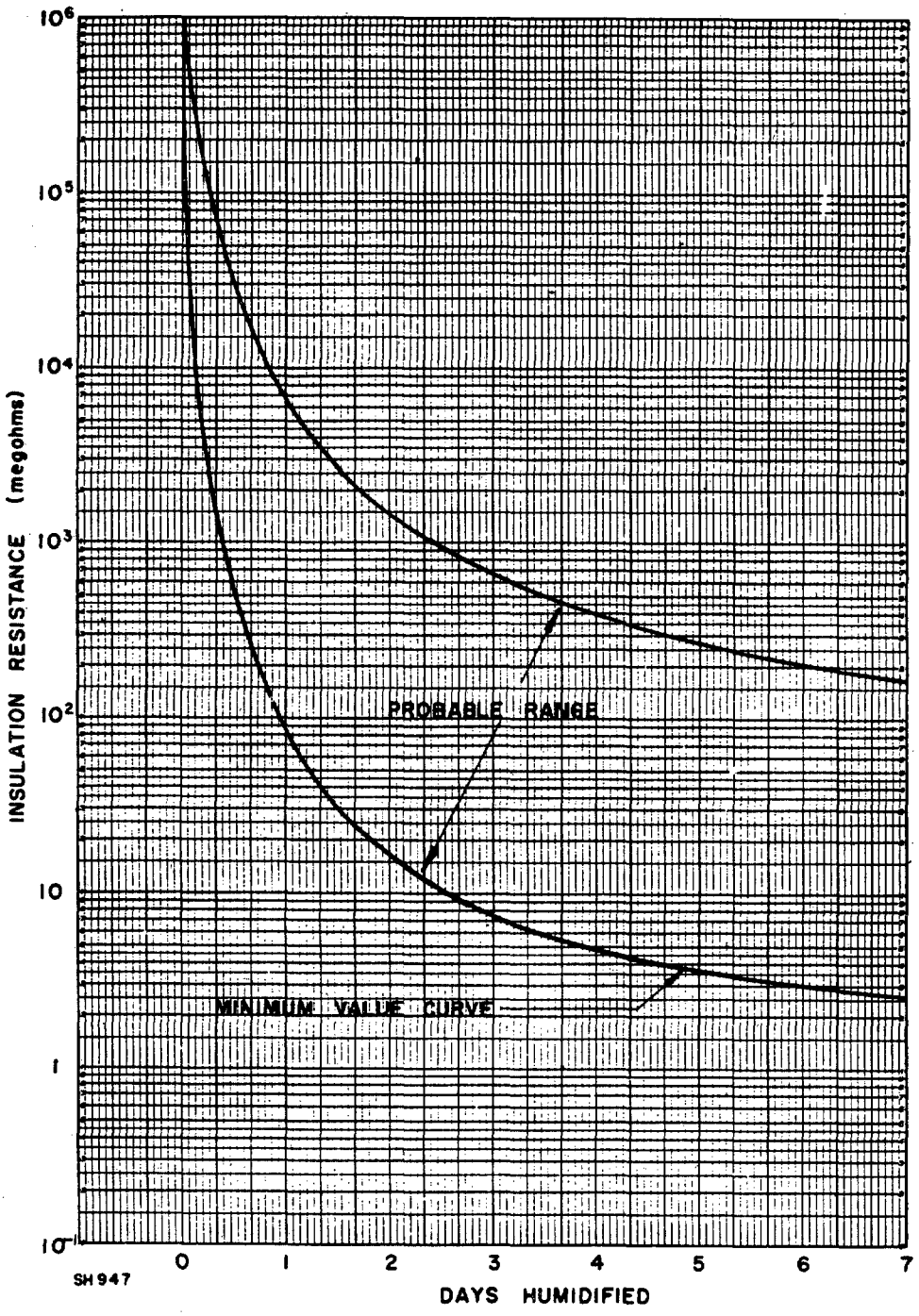


Figure 1 - Insulation resistance.

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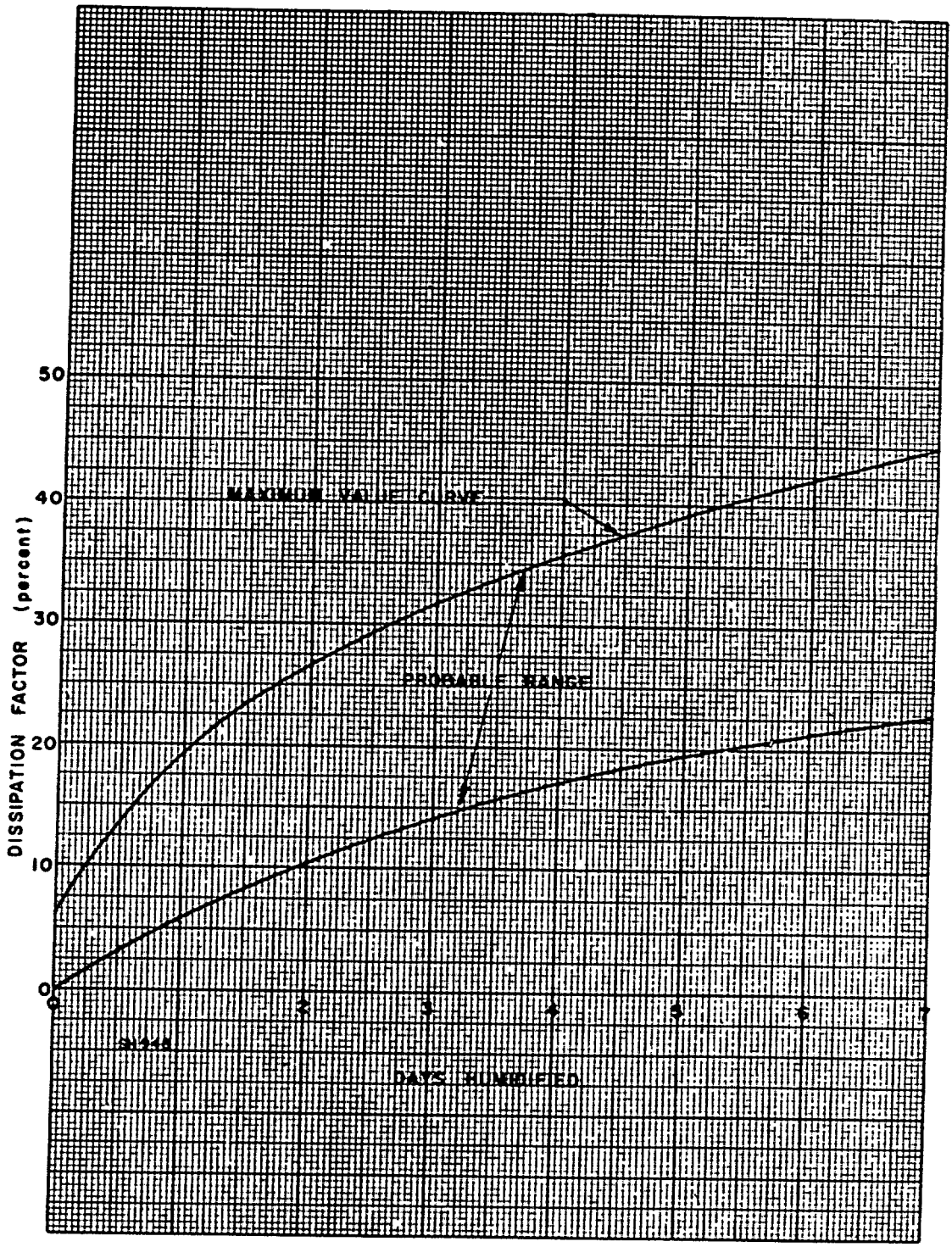


Figure 2 - Dissipation factor.

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